

VISHAAL S

✉ svishaalofficial@gmail.com

🌐 www.svishaal.vercel.app

📞 +91 6380520414

📍 Chennai, India

in <https://www.linkedin.com/in/svishaal/>

🔗 <https://github.com/Vishaal-sathya>

Actively learning, growing and seeking more challenges to expand my knowledge and experience so that I can positively impact and excel in any environment I am put into.

SKILLS

Agentic AI (CrewAI)	FastAPI (level 3 - HATEOAS)	Pytorch	Python
LLM & RAG Systems	DataBricks (PySpark)	PowerBI	SQL

EXPERIENCE

WALMART CENTER OF TECH EXCELLENCE - AI Engineer Intern

Jun 2025 - Aug 2025

- Worked on a project to identify emissions of harmful gases from micro, small and medium industries using satellite data.
- Trained a **contrastive learning model** on 7,000+ multi-spectral and hyper-spectral satellite images to infer industrial pollution levels, achieving a 87% accuracy in emission classification across test regions.

HYUNDAI MOTORS INDIA LTD - Data Analytics Intern

Oct 2024 - Apr 2025

- Conducted **supply chain risk assessment** through real time data analysis and designed interactive dashboards to gather insights.
- Created **audit automation systems** that helped to avoid repetitive work and optimize the project timeline.
- Built a project management system to improve day-to-day team efficiency and to provide actionable insights during meetings

PROJECTS

Communication System for Disaster Relief Operations

May 2024 - June 2024

- Developed a disaster-resilient communication network that transmits requests to relief agencies, when established communication and power systems are down.
- Implemented a **hybrid-search RAG system with re-ranking** for the supervision software that works with the communication system.
- The entire application helps Govt. officials to coordinate rescue and relief operations.

Iris Tracking Communication Aid for Paralyzed people

Mar 2024 - Mar 2024

- Developed a **OpenCV based Gaze detection** software for fully paralyzed individuals to communicate.
- It tracks eye movements, listens to conversations, and uses AI to generate responses, which are selected by eye movement and spoken aloud.

Autonomous Underwater Vehicle for Defect Detection in Dams

Nov 2023 - Dec 2023

- Autonomous underwater drone that detects and repairs cracks, corrosion, and abrasions in dam structures.
- Developed and integrated an **CNN based detector** for the drone which automatically detects cracks, rust and abrasions and also developed an **autonomous surveillance algorithm** for the drone.

KEY ACHIEVEMENTS & CERTIFICATES

- SAP Hackfest 2024** - 2nd place at national hackathon (prize: 1.5 lakhs)
- Smart India Hackathon** - Winner - 2025 (Prize : 1 Lakh), Finalist - 2024
- National Hackathon MIT** - Winner (Prize : 30000/-)
- IIT PALS INNOWAH National Hackathon** - Winner (Prize : 15000/-)
- Delegate** - Invited as delegate and exhibitor at Startup Mahakumbh 2024
- AWS Certified AI Practitioner**

EDUCATION

B.Tech Artificial Intelligence and Machine Learning

Nov 2022 - May 2026

Rajalakshmi Engineering College